Lab 4 实验报告

实验要求

CPU 需要支持以下指令并通过测试：

实现指令：`CSRRW`, `CSRRS`, `CSRRC`, `CSRRWI`, `CSRRSI`, `CSRRCI`

实现寄存器：`mstatus`, `mtvec`, `mip`, `mie`, `mscratch`, `mcause`, `mtval`, `mepc`, `mcycle`, `mhartid`, `satp`。这些寄存器均为64位宽。

实验过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

参考

- AI（DeepSeek）

Step 0

完成了 `csr_regfile.sv`，由 `AI` 完成。

Step 1

完成了对流水线的修改以及信号的连接。主要复用了 `alu` 模块和分支冲刷控制机制。

调试过程

这一小节中，所有 AI 帮助完成的部分都进行了标注

问题 1

由于 `csr` 寄存器的写回一开始放在了 `ex_stage`，导致 `csr` 的修改比提交早了两个周期。

解决 1

把写回放到了 `wb_stage`。并且发现，原本的分支冲刷逻辑依然可以复用，即只需要在 `ex_stage` 检测，冲刷 `if+stage` 和 `id_stage`。但是需要处理写后读逻辑。

问题 2

都是一些小问题，例如：写寄存器时，`MASK` 使用错误；`stall` 产生逻辑中没有考虑新加的 `csr` 读写指令。

以下是测试结果：

Lab 4

```
[src/cpu/cpu-exec.c:393,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x000000008001fff8
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 5,703 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 32,766
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 5,745,397 instr/s
```

Lab 3

```
AES benchmark + correctness
Running AES correctness checks...
[TEST] AES-128 test vector Encrypt
[PASS] AES-128 test vector Encrypt passed
[TEST] AES-128 test vector Decrypt
[PASS] AES-128 test vector Decrypt passed
[PASS] AES correctness checks passed
[RNG] heavy_rng_mod result=0, time(us)=255249
[RNG] heavy_rng_mod it/s=3921
[src/cpu/cpu-exec.c:393,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x0000000080000030
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 177,973 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 1,243,320
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 6,986,003 instr/s
```

Lab 2

```
[src/cpu/cpu-exec.c:393,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x000000008001fffc
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 5,643 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 32,767
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 5,806,663 instr/s
```

Lab 1

```
[src/cpu/cpu-exec.c:393,cpu_exec] nemu: HIT GOOD TRAP at pc = 0x0000000080010004
[src/cpu/cpu-exec.c:394,cpu_exec] trap code:0
[src/cpu/cpu-exec.c:74,monitor_statistic] host time spent = 2,617 us
[src/cpu/cpu-exec.c:76,monitor_statistic] total guest instructions = 16,385
[src/cpu/cpu-exec.c:77,monitor_statistic] simulation frequency = 6,260,985 instr/s
```